

CMPM 125 – Assignment 5

Game Expanded: Stealth Game

Team Members:

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Playable Link:

<https://smoopiz.itch.io/stealth-game>

Summary of Editions:

For our final project which built off of our assignment 4 stealth game, we built on the foundation that we had already created by adding more levels, stronger level progression, more complete UI, improved lighting and a little bit of narrative context.

One of the biggest editions to our game was the creation of 2 more complex floors that turned the map of the game into more of a building complex. These new floors added more space for the player to explore and made the gameplay more interesting since it gave the player more routes that they could use, more obstacles and more areas for guards to patrol to increase complexity.

We also set up guard routes on the new floors using our preexisting system for waypoints that we had originally created. This made it really simple to implement multiple new guards and increase difficulty by adding more chokepoints that would make it harder for the player to navigate on the upper levels. This made the guards feel more connected to the level design because their movement was no longer restricted to a simple area.

Another major edition that helped with the progression of the game was the keycard system. The player has to collect multiple keycards across a level in order to unlock access to the next floor. As the player ascends the floors, they will have to collect an increasing amount of keycards adding a sense of progression and difficulty so they don't just rush through the floors. On the top floor, we added the final win condition which was the item that their employer sent them to retrieve from the company so the player now had a clear goal as they progressed through the building.

We also improved the overall aesthetics of the game by creating a city level which would help create a more complete look for the game and serve as a shell for the inside levels that we created for a building. We also added better lighting in the building that would help give the building a more natural look with ceiling lights and dark corridors that would help build the stealth aesthetic.

In regards to the UI aspect, we created a main menu with a play button, credits screen, how to play screen and a quit button. We also designed it so it would have a more stealth game color scheme with a sort of dark navy blue. This helped the project

feel more complete and made it feel like a real game instead of just a prototype scene. We added instructions at the beginning of the game that the player has access to if they want to look at them so it can help them better understand the game's objective and controls before they start playing.

Another aspect of UI that we added was some narrative before the player starts playing the game. Once they press play the player is greeted with a UI panel with text that explains the premise of the game and what the goal of the player is. It explains to them everything they need to know from the elevator doors requiring keycards, the guards increasing in difficulty as they progress through the floors, and the item that they were sent to retrieve at the top of the building. This helped add a slight amount of narrative which helps tie the world together and make it so everything in the game world has a purpose.

Description of Graphics Integration from Assignment 3

For the graphics integration part of this assignment we chose to focus on two areas, atmospheric lighting and particle systems. For the lighting part we added more lights to every stage that we had along with roofs and removing the skybox to be able to control the lighting in every place in the stage so we could control the light distribution across the space. This allowed us to create a contrast between lighter and darker areas to reinforce the stealth aesthetic, and give the player hints about where to go or not to go. All of the lights we had used were kept white to help keep a realistic and grounded feel that would make a player believe that he's inside of a corporate building.

Next, for particle system we chose to use it as a way to give feedback to the player. Whenever a player picks up a keycard a sparkle burst particle effect plays, which gives the player a clear visual confirmation about the interaction. Along with that, we added a smoke screen particle effect that lets the player know when he uses his smoke bomb ability, which shows the player that the ability turned on and provides a reason in game to why the guards are losing sight of him.

List of Audio Assets/Other Assets

Background Music (4 tracks): DesiFreeMusic — Stealth Mission – Secret Spy Music, Stealth Mission – Secret Spy Music (Royalty Free), Stealth Mission Secret Spy Background Music for Covert Ops, Perfect for Stealth Sniper Missions Music

Keycard Pickup SFX: Mellow Fox Studios / litupsubway — City Level Building Assets - Key Collect SFX

Access Denied SFX: miraclei via DRAGON-STUDIO — sample_error_fail01

Smoke Bomb SFX: vckhaze — via Freesound

Jump Landing SFX: meggiepie — via Freesound

Footstep SFX: FreeSoundsxx — Walking on Concrete ver 2

**Guard Voice Lines: SoundBiter — NPC Guards Warning Phrases Audio Pack
(voicebosch.itch.io)**

Smoke VFX Pack: Maiami Studio — Free Stylized Smoke Effects Pack

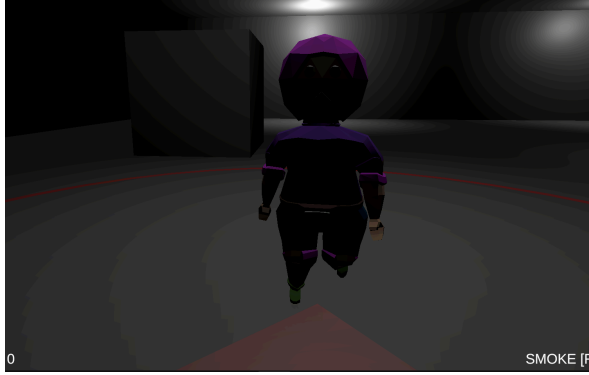
Guard Character Model: ResilientLogic — Chuby Character LowPoly Free

Final Room Chest Model: Dragan Maric Gale — Chests Fantasy PBR - Mobile

Screenshots Showcasing Key Features



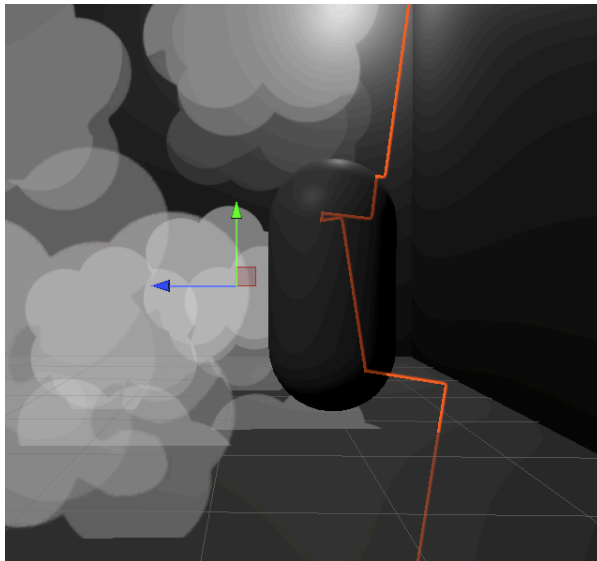
Main Menu



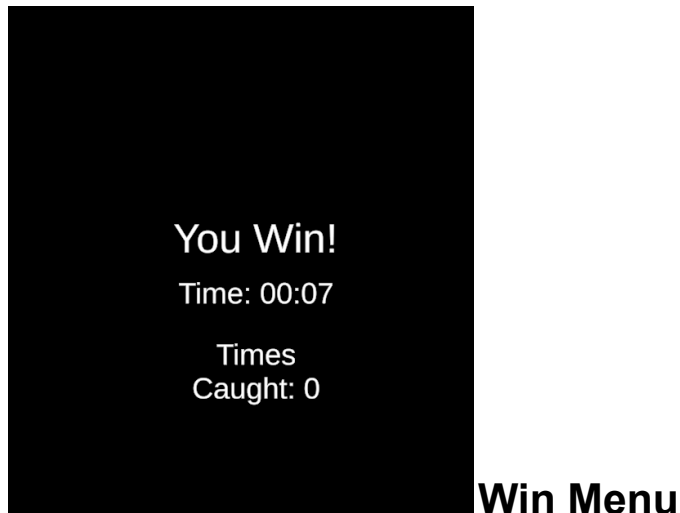
Guard Chasing the Player



Lose Menu



Smoke Screen Ability



Team Reflections:

Yahir Rico:

For this project, I worked on creating the main menu which had the play, credits, how to play, and quit buttons along with the contents of the how to play. I also added the instructions at the beginning of the game which help give the player a better understanding of what they need to do before playing the game.

I also contributed to the game's level design where I created the city level which would help give our game a better environment and serve as a shell for the levels that we created so we could focus on the interior level design. I created the two new more complex floors and set up the routes for the guards across the levels. This made the stealth aspect of the game a lot more interesting because the player has to move through much more complex levels while having to pay attention to the increasing amount of guards as they progress. I also worked on the keycard functionality which has the player collect keycards throughout the floor and if they have the required amount they are transported to the next level of the building.

I also added the win condition where if they reached the top floor as instructed and picked up the target item, they would win the game. This helped give the game a clear progression through having to collect an increasing amount of keycards and dealing with an increasing amount of guards at the same time as they move onto the upper floors.

One challenge I had was making sure that all of these new systems would work together correctly. One issue was with the elevator which for some reason wouldn't transport the player even though all the conditions were correct and would just shake the camera as if it actually teleported. I had to refine the teleport system which simulated the elevator to have better collisions so it wouldn't have any problems transporting the player.

Through this project, I also learned more about organizing the unity hierarchy and scene to make it much easier to work on. Specifically, I made it so I would separate the different levels in the hierarchy so I can hide the other floors so there would be less to focus on. This would also make it much easier to add and remove things while also testing new features.

Yuval Szwarcbord:

For this project I was responsible for programming, assets, and UI for the game. On the programming side, I had built and worked on multiple of our scripts that connect to each other, specifically guard AI, which features a state machine with multiple states, a vision cone, and a hearing cone, as well as audio cues that'll tell the player whatever state the guard is in. I also developed the keycard script further to allow a particle system and audio system to work together with it for pickup. Then using the UI aspect, I added how many keycards are needed to move onto the next floor so the player will be able to reach the elevator for the next door without having to worry about whether they have enough keycards. On top of that I had added an audio setting system with persistent volume controls across different scenes, and a music manager that does the same. Then I added a UI system that displays the timer, how many times the player got caught, the keycard progress, and the smoke screen ability.

One of the bigger challenges I tackled was building a checkpoint and reset system that avoids reloading the scene entirely. With the original system we had, whenever the player got caught the whole scene would be reloaded since the original stealth game was one floor, which wouldn't work for this upgraded game. We had to rework our systems to allow multiple floors, with respawning, resetting keys, and guards. Getting this to work correctly across multiple floors in the same scene required a tag-based floor configuration system so the right respawn point triggers depending on which floor's guard catches the player. For the UI I designed and implemented the in-game HUD, the audio settings menu, and the win screen which displays the player's final time and how many times they were caught.

One of the main things I learned through this project was how quickly small systems grow in complexity once they need to interact with each other. Something like the respawn system sounds simple until you realize it needs to talk to the guard AI, the keycard manager, the loss curtain, the HUD and the player controller all at once. If I had more time I would've liked to make more guard types with different abilities, and polish the existing guards even further, add more abilities for the player and a way to unlock them, add more floors, and finally refine the difficulty curve further.